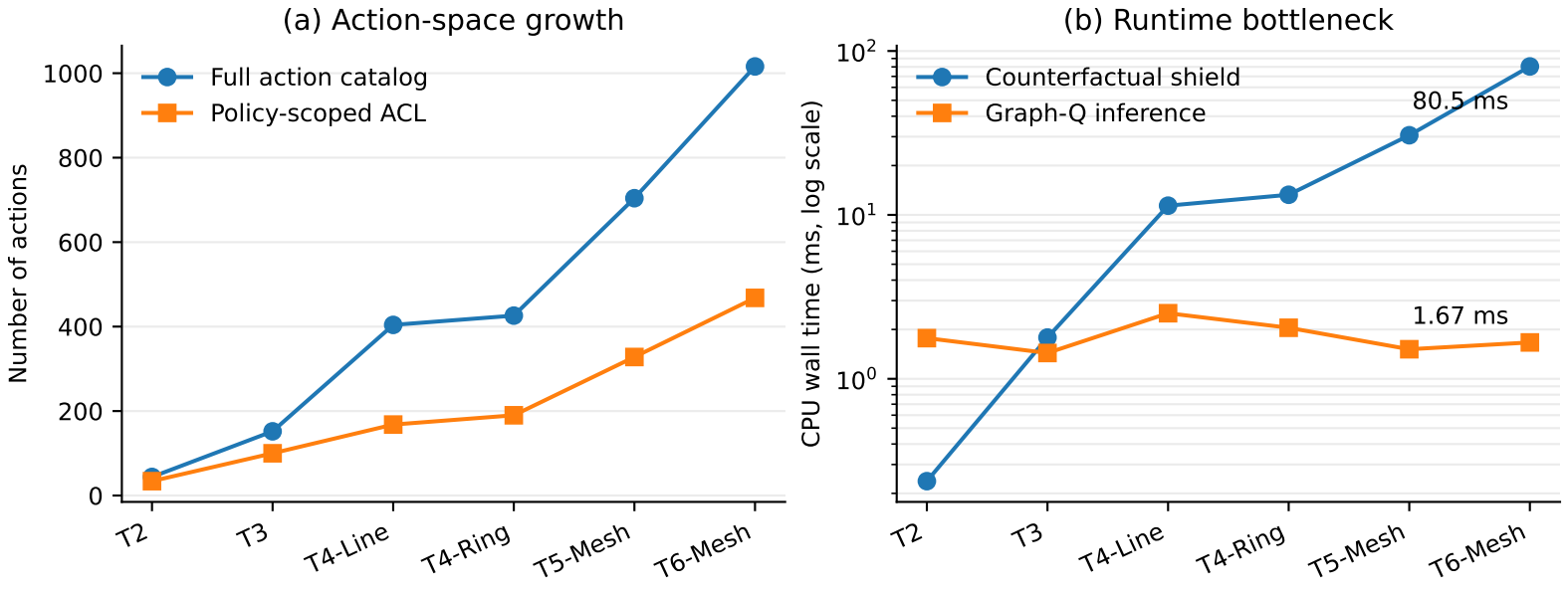
(a) Action-space growth
Number of actions
0
200
400
600
800
1000
Full action catalog
Policy-scoped ACL
T2
T3
T4-Line
T4-Ring
T5-Mesh
T6-Mesh
(b) Runtime bottleneck
CPU wall time (ms, log scale)
10^0
10^1
10^2
Counterfactual shield
Graph-Q inference
80.5 ms
1.67 ms
T2
T3
T4-Line
T4-Ring
T5-Mesh
T6-Mesh